TASK 4: 24V ELECTRICAL MEASUREMENT AND TECHNICAL DOCUMENTATION

Heavy Vehicle Service and Maintenance Technician Level 3 | Block 1 | v1.1

TIME	EVIDENCE	PROGRAMME POINT
90 minutes (adapt to workplace conditions)	**Practical | Knowledge | Documentation**	**Weeks 1-10 | Months 1-3**

Your task

Use technical information and a digital multimeter to check a protected 24 V training circuit or heavy-vehicle low-voltage system, then complete a clear technical record and handover.

What you must do

1. Identify circuit hazards, voltage level, isolation needs and safe test points.
2. Select and prove the meter, leads and range before connection.
3. Measure supply voltage, voltage drop and resistance only where the procedure permits.
4. Interpret results against the circuit diagram and specification.
5. Record findings, reinstate protection and communicate the result.

What good evidence looks like

SAFE CONTROL

Follow manufacturer isolation procedures; never bridge circuit protection or probe connectors in a way that causes damage.

CORRECT METHOD

A logical sequence using current manufacturer, DVSA, company and task-specific information.

ACCURATE EVIDENCE

Actual values, test conditions, units, findings and evidence references recorded clearly.

SOUND DECISIONS

A justified serviceability, repair, referral or recovery decision followed by appropriate verification.

You will need

- 24 V training board or heavy vehicle
- Digital multimeter and fused leads
- Wiring diagram and specifications
- PPE and job-card documentation

SOW ALIGNMENT

Block 1 - Motor Vehicle Foundation Skills

KSB evidence focus: Controlled Foundation Skills mapping; primary heavy-vehicle link K4

Task reference: HV_L3_Block_1_Task_4_24V_Electrical_Measurement_and_Technical_Documentation

TASK 4: 24V ELECTRICAL MEASUREMENT AND TECHNICAL DOCUMENTATION

Practical record

Apprentice, location and vehicle details

Apprentice name

Date

Employer / training centre

Vehicle / trailer make and model

Registration / fleet number

Odometer / hours

VIN / chassis / trailer number

Job card / task reference

Measurements and technical checks

Open-circuit voltage		Loaded voltage	
Positive-side voltage drop		Ground-side voltage drop	
Resistance / continuity		Fuse / protection rating	
Specified range		Conclusion / circuit state	

Special tools, equipment, calibration or technical-information requirements

Task evidence

PPE, precautions and safe-system controls used

Safety-critical findings and immediate action

Work completed, actual findings, diagnosis, recommendations and verification

Knowledge check

1. Why is resistance not measured on an energised circuit?

2. Why should a meter and its leads be proved before and after tes

Knowledge check - continued

3. Explain how voltage-drop testing differs from continuity testing.

4. What information must a useful electrical job card contain?

5. When should an electrical concern be isolated and referred?

Assessor/Workplace Mentor observation

- ☐ Identifies voltage, stored-energy and short-circuit risks.
- ☐ Selects correct meter function, range and connection method.
- ☐ Uses a circuit diagram to choose meaningful test points.
- ☐ Records units, conditions and results accurately.
- ☐ Reinstates covers, protection and vehicle state.

Assessor/Workplace Mentor observation notes

Decision and feedback

Decision - select one

- ☐ PASS
- ☐ REFER

Evidence reference

Feedback and action required

Sign-off

Signatures confirm that this is the apprentice's own work and that the evidence and developmental decision are accurate.

Apprentice name

Assessor/Workplace Mentor name

Apprentice signature

Assessor/Workplace Mentor signature

Date

Heavy Vehicle Level 3

Task answer continuation | Form accessibility revision 12 November 2026

Use this page when an answer will not fit visibly in its original box.
Reference the question and write "See continuation page" in the original answer.
Keep each response within the visible space. Add another copy of this page if needed.

Apprentice name

Block / task / page number

Question / section reference

Continued answer

Question / section reference

Continued answer

Continuation evidence must be reviewed with the original task and referenced at sign-off.